(Please write your Enrollment Number) Enrollment No._____

REMID-TERM EXAMINATION
(Course Name: B.Tech) (Semester: 7th)
(November 2024) OFF LINE mode

Subject Code: BAI 403	**Subject: Computer Vision**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**	**CO Mapping**
a)	What do you mean by digitization of image?	2.5	1
b)	Write and explain various applications of computer vision?	2.5	1
c)	Briefly explain edge detection.	2.5	1
d)	How CNN differs from ANN?	2.5	1
	UNIT I		
Q2	**Attempt any two parts**	**(5*2=10)**	**CO Mapping**
a)	Describe the process of gradient averaging in SPGD. How does it affect the convergence and stability of the algorithm?	05	1
b)	Explain the concept of backpropagation and its role in training neural networks. Why is it essential for learning in deep networks?	05	1
c)	Explain the concept of linear classification. How does a linear classifier separate data into different classes?	05	1
	UNIT II		
Q3	**Attempt any two parts**	**(5*2=10)**	**CO Mapping**
a)	What is the function of ReLU layer. Give suitable example.	05	1
b)	What are the different types of pooling? Explain their characteristics.	05	1
c)	Write short notes on i) LeNet ii) ResNet	05	1